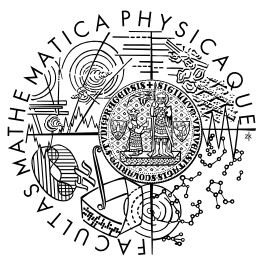


Charles University
Faculty of Mathematics and Physics

BACHELOR THESIS

Onsager Machlup Principle and Machine Learning



**FACULTY
OF MATHEMATICS
AND PHYSICS**
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BACHELOR THESIS

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Onsager Machlup Principle and Machine Learning

Mathematical Institute of Charles University

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Study programme: Mathematical Modelling

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Title: Onsager Machlup Principle and Machine Learning

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Abstract:

We study the most probable path in configuration space for overdamped Langevin dynamics, given by the minimizer of the Onsager Machlup action, which we find using neural networks. From the most probable path we calculate the free energy profile, transition states and reaction time without having to perform expensive MD simulations. We develop a simple method where continuity and boundary conditions are automatically satisfied without adding extra terms to the action. We implement the method in PyTorch. We then compute how the action depends on the total time in accordance with Hamilton-Jacobi theory.

Keywords: Onsager Machlup action, neural networks, transition state theory

Název práce: Onsager Machlup princip a strojové učení

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Abstrakt:

Zabýváme se nejpravděpodobnější trajektorií v konfiguračním prostoru pro přetlumenou Langevinovu dynamiku. Nejpravděpodobnější trajektorie je minimizér Onsager Machlupovy akce, který hledáme pomocí neuronových sítí. Počítáme z něj profil volné energie a přechodový stav, aniž bychom museli provádět náročné MD simulace. V naší metodě spojitost trajektorie a okrajové podmínky vychází automaticky bez přidání členů do akce. Metodu implementujeme v PyTorch. Poté počítáme, jak akce závisí na celkovém čase v souladu s Hamilton-Jacobiho teorií.

Klíčová slova: Onsager Machlupova akce, neuronové sítě, teorie přechodného stavu

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List of Abbreviations

OM	Onsager Machlup action
FW	Friedlin Wentzel action
MEP	Minimum Free Energy Path
NEB	Nudged Elastic Band method
MD	Molecular dynamics
NN	Neural Network
MB	Müller Brown potential
LJ	Lennard Jones potential
LJ ₇	Lennard Jones clusted with 7 atoms
MFPT	Mean First Passage Time

Introduction

Chemical kinetics 1 is a subfield of physical chemistry that tries to understand the rates of reactions, in particular how these rates depend on concentration, temperature, pressure, presence of catalysts and other external properties. Transition state theory 2 tries to explain these rates by considering the transition state: the configuration that the particles achieve during the reaction, which has the highest energy. The difference between energy of the transition state and energy of reactants is called the activation energy 3, it is the smallest amount of energy the system needs for the reaction to happen. More detailed information comes with the reaction coordinate 4, which is a time-like coordinate representing the progress of the reaction. In particular one is interested in how the free energy depends on the reaction coordinate, a so called free energy profile 5.

If we knew the detailed evolution of all positions and velocities of particles during the reaction, we could compute all of these quantities of interest. In this work, we propose a way to do this based on the Onsager Machlup (OM) action 6.

The chemical reaction can be modeled as a stochastic process within a potential landscape. The state of the system is described by a process $\mathbf{x}_t$ which evolves according to the overdamped Langevin equation 7

$$d\mathbf{x}_t = -\zeta^{-1}\nabla U(\mathbf{x})dt + \sqrt{2D}d\mathbf{W}_t,$$

where ζ is the friction coefficient, $U(\mathbf{x})$ is the potential energy landscape, D is the diffusion constant, and $d\mathbf{W}_t$ is an increment of the Wiener process.

The most probable path $\mathbf{x}(t)$ connecting two points $\mathbf{x}_0$ and $\mathbf{x}_f$ in time T is given by the minimizer of Onsager Machlup action:

$$S[\mathbf{x}(t)] = \frac{1}{\zeta}(U(\mathbf{x}_f) - U(\mathbf{x}_0)) + \frac{1}{2} \int_0^T \left(\dot{\mathbf{x}}^2 + \left(\frac{\nabla U}{\zeta} \right)^2 - \frac{2D}{\zeta} \Delta U \right) dt \quad (1)$$

Unlike previous research done with the OM action 8, we encode the path in a Neural Network (NN) 9 as a continuous function $\mathbf{x}: [0, 1] \rightarrow \mathbb{R}^n$, where the time input is one neuron and the space output is N neurons. We only discretize to compute integrals and derivatives. Therefore we do not need to enforce continuity or even boundary conditions by adding extra terms. From the most probable path, we compute free energy profiles, transition states, activation energy.

- In chapter 1, we lay out the numerical methods used, and explain how we check convergence of NN to the minimizer via conservation of Hamiltonian.
- In chapter 2, we illustrate the method on the Müller Brown and Lennard Jones potentials.
- In chapter 3, we explain (an unsuccessful) method for estimating the Mean First Passage Time.

The contributions of author are: incoding the path as a continuous function with affine transformation, implementation in PyTorch, calculating dependence of minimizer on total reaction time.

1 Theory and Methods

In this chapter, we explain some properties of the OM action and how we compute it's minimizer and discuss conservation of Hamiltonian and saddle points. In the following chapter 2, the results of our methods are shown for the Müller Brown and Lennard Jones potentials.

1.1 Meaning of the Onsager Machlup action

Consider a particle undergoing overdamped Langevin dynamics

$$d\mathbf{x}_t = -\zeta^{-1}\nabla U(\mathbf{x})dt + \sqrt{2D}d\mathbf{W}_t,$$

Then given the particle at time 0 is known to be at $\mathbf{x}_0$, the probability density of finding the particle at $\mathbf{x}$ at time t is $\Pi(\mathbf{x}, t; \mathbf{x}_0, 0)$, where Π satisfies the Fokker Planck equation (10, chapter 4):

$$\partial_t \Pi - \nabla \cdot \left(\Pi \frac{\nabla U}{\zeta} \right) - D \Delta \Pi = 0 \quad (1.1)$$

Eq (1.1) has a stationary solution, the Maxwell-Boltzmann distribution $\Pi(\mathbf{x}) = \exp(-\frac{U(\mathbf{x})}{k\theta})$, where θ is temperature of the environment. By plugging it into Eq (1.1), we get $D\zeta = kT$, the Einstein Stokes relation (10, section 3.1). The solution of Eq (1.1) can be written as a path integral

$$\Pi(\mathbf{x}, t; \mathbf{x}_0, 0) = \int_{\mathbf{x}_0, 0}^{\mathbf{x}, t} \exp \left(-\frac{S[\mathbf{x}(t)]}{2D} \right) \mathcal{D}\mathbf{x}(t),$$

where S is the OM action (see Appendix A, or 10, chapter 6). We see that the paths with the highest contribution to Π are the ones with smallest OM action, and this is why we define the most probable path to be $\arg \min S[\mathbf{x}]$.

There are other methods to calculate transition paths, like Freidlin-Wentzel (FW) 11 or Minimum Free Energy Path (MEP) 12. FW is the Onsager-Machlup action without ΔU term, which can be thought of as a zero diffusion (equivalently zero temperature, as $D\zeta = k\theta$) limit of OM. Actually, when interpreted as Itô integrals acting on non-differentiable paths $\mathbf{x}$ (solutions of the underlying SDE), the OM and FW actions lead to the same result, due to non-standard calculus for Itô integrals. However, they are different for C^1 paths, and for finding the most probable path, OM is indeed the correct choice (for more on this delicate matter see 13).

Moreover, in the FW limit $D \rightarrow 0$, a large deviation principle holds:

$$\mathbb{P} \left[\sup_{t \in [0, T]} |\mathbf{x}_t - \psi(t)| < \delta \right] \sim \exp \left(\frac{-S_{FW}[\psi]}{D} \right),$$

meaning that most trajectories of $\mathbf{x}_t$ lie in a small tubus around $\arg \min S[\mathbf{x}]$. 14

The MEP minimizes the line integral $\int_{\mathbf{x}} |\nabla U(\mathbf{x})| ds$, it is the limit of FW paths (up to reparametrization) as the final time goes to ∞ and therefore not well suited for analysis of finite-time events. To calculate the MEP, string methods are often used, where the path is discretized by finitely many points. To keep the nodes of string at equal distance, one needs to add additional spring force, this is called the Nudged elastic band (NEB) method. There are yet other methods, see 15.

1.2 Neural networks

We encode the path by a Multilayer perceptron 9. This is a simple type of neural network where neurons are organized in layers and the computation follows from layer i to layer $i + 1$ via affine transformation followed by nonlinear activation functions. The purpose of activation functions a_1, a_2, a_3 is to generate a class of functions rich enough such that every continuous function can be approximated by such compositions 16. In our case the first (input) layer is 1D, corresponding to time, and the final layer is nD, corresponding to a point in configuration space $\mathbb{R}^n$. So our NN looks like (N_1, N_2 are number of neurons in hidden layers)

$$\mathbf{x}_{NN}(t) = \mathbb{A}_3 a_3 (\mathbb{A}_2 a_2 (\mathbb{A}_1 t + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3$$

$$\mathbb{A}_1 \in \mathbb{R}^{1 \times N_1}, \mathbf{b}_1 \in \mathbb{R}^{N_1}, \mathbb{A}_2 \in \mathbb{R}^{N_1 \times N_2}, \mathbf{b}_2 \in \mathbb{R}^{N_2}, \mathbb{A}_3 \in \mathbb{R}^{N_2 \times n}, \mathbf{b}_3 \in \mathbb{R}^n$$

Neural networks are trained by a gradient descent algorithm minimizing the loss function $\mathcal{L}[\mathbf{x}]$, the concrete details of descent given by an optimizer (we use Adam 17). We use the OM action itself as a loss function. Unlike loss functions in machine learning, it is not computed from golden data, it does not have to be non-negative nor convex.

1.3 Formulation of numerical method

We return to Eq (1). The term $U(\mathbf{x}_f) - U(\mathbf{x}_0)$ can be dropped since it does not effect the minimization, and changing to dimensionless variables $\hat{U}, \hat{x}, \hat{t}$ with characteristic scales $\bar{U}, \bar{x}, \bar{t}$ satisfying $U = \bar{U}\hat{U}$, $x = \bar{x}\hat{x}$, $t = \bar{t}\hat{t}$, $\hat{t} \in (0, 1)$, we get

$$S = \int_0^{\hat{t}} \dot{\mathbf{x}}^2 + \left(\frac{\nabla U}{\zeta} \right)^2 - \frac{2D}{\zeta} \Delta U dt = \frac{\bar{x}^2}{\bar{t}} \int_0^1 (\dot{\hat{\mathbf{x}}})^2 + \alpha (\nabla_{\hat{\mathbf{x}}} \hat{U})^2 + \beta \Delta_{\hat{\mathbf{x}}} \hat{U} d\hat{t}$$

$$\alpha = \frac{\bar{U}^2 \bar{t}^2}{\zeta^2 \bar{x}^4}, \quad \beta = -\frac{2D \bar{U} \bar{t}^2}{\zeta \bar{x}^4}$$

and from now on minimize $\hat{S}$ depending on α, β . Here β is a dimensionless measure of diffusion, the limit $\beta = 0$ corresponds to FW action.

So far our $\mathbf{x}_{NN}$ does not have fixed endpoints. To achieve this, we need to find an affine transformation matrix that sends $\mathbf{x}_{NN}(0) \mapsto \mathbf{x}_0$, $\mathbf{x}_{NN}(1) \mapsto \mathbf{x}_f$. For details see Algorithm 1.

The path thus obtained is continuous and we only discretize to compute (1). We have to choose nodes $\{t_i\}_{i=1}^N$ in time and evaluate $\mathbf{x}_{NN}(t_i)$ to compute $\dot{\mathbf{x}}$ and the integrals. Space derivatives $\nabla U, \Delta U$ are computed with automatic differentiation, which is also used in the gradient descend itself to differentiate $\nabla_{\mathbb{A}_i} \mathcal{L}, \nabla_{\mathbf{b}_i} \mathcal{L}$. The training of $\mathbf{x}$ is given in Algorithm 2. The advantage of having a continuous path to begin with is we don't have to add any spring terms, or interpolate to increase the number of nodes in time.

Training of the NN was implemented in PyTorch 18.

Algorithm 1 Affine transformation to force initial and final condition

Input: $\mathbf{x}_0, \mathbf{x}_f, \mathbf{x}_0^{\text{desired}}, \mathbf{x}_f^{\text{desired}} \in \mathbb{R}^d, \mathbf{x} \in \mathbb{R}^d$
Output: $T(\mathbf{x})$ such that T affine, $T(\mathbf{x}_0) = \mathbf{x}_0^{\text{desired}}, T(\mathbf{x}_f) = \mathbf{x}_f^{\text{desired}}$
 $v \leftarrow \mathbf{x}_f - \mathbf{x}_0$
 $v_{\text{desired}} \leftarrow \mathbf{x}_f^{\text{desired}} - \mathbf{x}_0^{\text{desired}}$
 $\text{scale} \leftarrow \frac{\|v_{\text{desired}}\|}{\|v\|}$
 $v \leftarrow \text{scale} \cdot v$
 $q \leftarrow v_{\text{desired}} - v$
 $q \leftarrow \frac{q}{\|q\|}$
 $H \leftarrow I - 2qq^T$
 $T(\mathbf{x}) \leftarrow \mathbf{x} - \mathbf{x}_0$
 $T(\mathbf{x}) \leftarrow \text{scale} \cdot T(\mathbf{x})$
 $T(\mathbf{x}) \leftarrow H \cdot T(\mathbf{x})$
 $T(\mathbf{x}) \leftarrow T(\mathbf{x}) + \mathbf{x}_0^{\text{desired}}$
return $T(\mathbf{x})$

Algorithm 2 Training of NN

Input: potential $U: \mathbb{R}^d \rightarrow \mathbb{R}$, constants $D, \zeta > 0, \bar{\mathbf{x}}, \bar{t}, \bar{U}$, NN architecture $\mathbf{x}_{NN}(t; \theta)$ for parameters $\theta = (\mathbb{A}_i, \mathbf{b}_i)$, learning rate lr , optimizer, number of epochs n , number of nodes in time $N = \frac{1}{\Delta t}$
Output: $\theta^* = \arg \min_{\theta} S[T(\mathbf{x}(\bullet; \theta))]$
Initialize θ_0 randomly
for epoch = 1, ..., n **do**
 $\mathbf{x}_i \leftarrow \mathbf{x}_{NN}(t_i; \theta)$ for all i
 $\mathbf{x}_i \leftarrow T(\mathbf{x}_i)$
 Calculate $\nabla U(\mathbf{x}_i)$ and $\Delta U(\mathbf{x}_i)$ via autodiff
 $S(\theta) \leftarrow S[\{\mathbf{x}_i\}_{i=1}^N] = \sum_i \frac{(\mathbf{x}_{i+1} - \mathbf{x}_i)^2}{\Delta t} + \alpha(\nabla U(\mathbf{x}_i))^2 \Delta t + \beta \Delta U(\mathbf{x}_i) \Delta t$
 $\nabla_{\theta} S$ via backpropagation
 Update parameters:
 $\theta \leftarrow \text{Adam}(\theta, \nabla_{\theta} S, lr)$
end for

1.4 Correctness check: Hamiltonian

In machine learning, the loss function used is often L^2 distance between generated data and golden data, so the closer it is to zero, the better fit we have. Since the loss function we are using does not (necessarily) have minimum 0 and we do not know à priori it's minimum, we must check convergence of training by other methods.

We are minimizing the integral of a function L depending on $(\mathbf{x}, \dot{\mathbf{x}}) \in \mathcal{T}\mathbb{R}^N$,

$$S[\mathbf{x}] = \int_0^1 L(\mathbf{x}, \dot{\mathbf{x}}) dt, \quad L(\mathbf{x}, \dot{\mathbf{x}}) = \dot{\mathbf{x}}^2 + \alpha(\nabla U(\mathbf{x}))^2 + \beta \Delta U(\mathbf{x}).$$

Therefore the *Hamiltonian* H , the legendre transform of L , is conserved 19:

$$H(\mathbf{x}, \dot{\mathbf{x}}) = \dot{\mathbf{x}}^2 - \alpha(\nabla U(\mathbf{x}))^2 - \beta \Delta U(\mathbf{x}) = \text{const.}$$

In the numerical experiment shown in chapter 2, the Hamiltonian is always conserved to good accuracy, which we measure as $\frac{\sigma_H}{\overline{H}} = \frac{\sqrt{\sum_{i=0}^{n-1} (H(t_i) - \overline{H})^2}}{\overline{H}}$.

1.5 Saddle points

In the chapter 2 we show both for MB and LJ potentials that in the FW limit $D \rightarrow 0$, the minimizer passes through a saddle point $\mathbf{x}_{saddle}$ of U . This result is important for transition state theory, because it means that in the $D \rightarrow 0$ regime we can equate saddle points and transition points. From the point of view of chemical kinetics, the activation energy of a reaction is $U(\mathbf{x}_{transition}) - U(\mathbf{x}_0) = U(\mathbf{x}_{saddle}) - U(\mathbf{x}_0)$. We will see importance of activation energy in chapter 3.

This behaviour is explained in 20, chapter 4, section 2, theorem 2.1, where the following is proved:

Theorem 1 (Low noise langevin process escapes from saddle). *Let $\mathbf{x}^D, \mathbf{x}$ solve*

$$\begin{aligned} d\mathbf{x}_t^D &= -\zeta^{-1} \nabla U(\mathbf{x}_t^D) + \sqrt{2D} d\mathbf{W}_t \\ \dot{\mathbf{x}} &= -\zeta^{-1} \nabla U(\mathbf{x}) \end{aligned} \tag{1.2}$$

Let $\mathbf{x}_0$ be a minimum of $U(\mathbf{x})$ that is attractor of system (1.2), Ω a domain containing $\mathbf{x}_0$,

$$\mathbf{x}_{saddle} := \arg \min_{\mathbf{y} \in \partial\Omega} U(\mathbf{y}),$$

and $\tau^D := \inf\{t \geq 0 : \mathbf{x}_t^D \notin \Omega\}$ be first escape time of $\mathbf{x}^D$ from Ω . Then

$$\forall \delta > 0 : \lim_{D \rightarrow 0^+} \mathbb{P}[\|\mathbf{x}_{\tau^D}^D - \mathbf{x}_{saddle}\| < \delta] = 1$$

2 Examples

In this chapter we show the results of our trained Neural Networks on the Müller Brown (MB) and Lennard Jones (LJ) potentials.

2.1 Müller Brown Potential

The MB potential²¹ is a two-dimensional analytical potential given by

$$U(x, y) = \sum_{i=1}^4 A_i \exp(a_i(x - \bar{x}_i)^2 + b_i(x - \bar{x}_i)(y - \bar{y}_i) + c_i(y - \bar{y}_i)^2)$$

for certain constants A_i, a_i, b_i, c_i . It has in total three minima and two saddle points, we are concerned with two minima and one saddle as shown on Figure 2.1. The MB potential is a good toy model for testing transition path sampling algorithms, because a straight path connecting the two minima is not a good path, in the sense of OM or FW or MEP: its gradient $|\nabla U|^2$ is too high.

The minimizing path $\mathbf{x}(t) = (x(t), y(t))$ is found for cases $\beta = 0$ (zero diffusion = zero temperature limit, since $D\zeta = k\theta$) and $\beta \neq 0$. Results are shown on Figure 2.1.

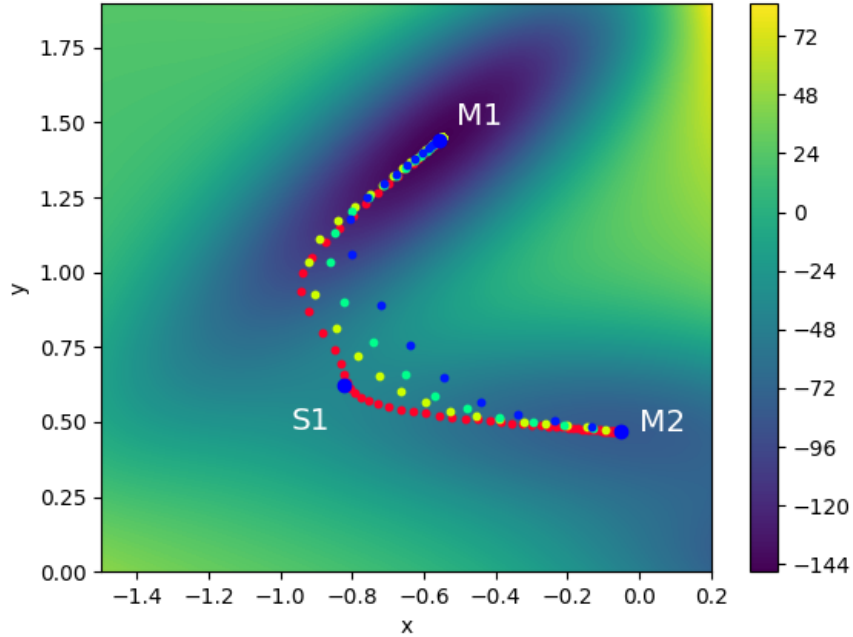


Figure 2.1 Contour plot of MB potential with minima M1, M2 and saddle S1. We are considering paths starting at $\mathbf{x}_0 = \text{M1}$ and ending at $\mathbf{x}_f = \text{M2}$. We plot two minimizers of OM action for values $\alpha = 0.01, \beta = 0$ (red), $\alpha = 0.01, \beta = 0.05$ (yellow), $\alpha = 0.01, \beta = 0.1$ (green), $\alpha = 0.01, \beta = 0.15$ (blue). Only 200 points $\mathbf{x}(t_i), t_i = \frac{i}{200}, i \in \{0, \dots, 199\}$ on the path are plotted to visualize particle velocity, which is slower near the minima. Still underneath, the path is continuous $\mathbf{x}: [0, T] \rightarrow \mathbb{R}^N$. Training converged after 2000 epochs with two hidden layers of 16 neurons, 1000 points in time discretization.

We can see that in the low temperature limit, the solution passes through a saddle, as predicted by 20. The higher the temperature, the more the path will "skip".

2.2 Lennard Jones Potential

The LJ potential 22 is an interatomic potential used to model interaction between rare gas atoms. It consists of a short range repulsive r^{-12} and long range attractive r^{-6} terms. It is perhaps the most studied potential in computational chemistry and statistical mechanics, because it is simple yet has many interesting properties: for example it can be in solid, liquid, gas and supercritical phase depending on temperature and density, and one can calculate the virial coefficients by hand. Moreover, finding all minima and saddles for various LJ_n clusters is a benchmark for various minimization algorithms, see 23. We consider a 7 atom cluster (LJ_7), in which case the configuration space has $7 \times 3 = 21$ dimensions:

$$\mathbf{x} = (\underbrace{x_1, x_2, x_3}_{\text{1st atom}}, \underbrace{x_4, x_5, x_6}_{\text{2nd atom}}, \dots, x_{21})$$

$$r_{ij} := \sqrt{((x_{3i+1} - x_{3j+1})^2 + (x_{3i+2} - x_{3j+2})^2 + (x_{3i+3} - x_{3j+3})^2)}$$

$$U(\mathbf{x}) = 4\epsilon \left[\sum_{1 \leq i < j \leq 7} \left(\frac{r_0}{r_{ij}} \right)^{12} - \left(\frac{r_0}{r_{ij}} \right)^6 \right]$$

Where the constants ϵ, r_0 determines energy of interaction and characteristic distance (in case of two atoms $U_{min} = U(r = 2^{\frac{1}{6}}r_0) = -\epsilon$). It is well known 24 that LJ_7 has 4 minima and 3 saddles. We focus on the two lowest energy minima, which can be seen on Figures 2.2:

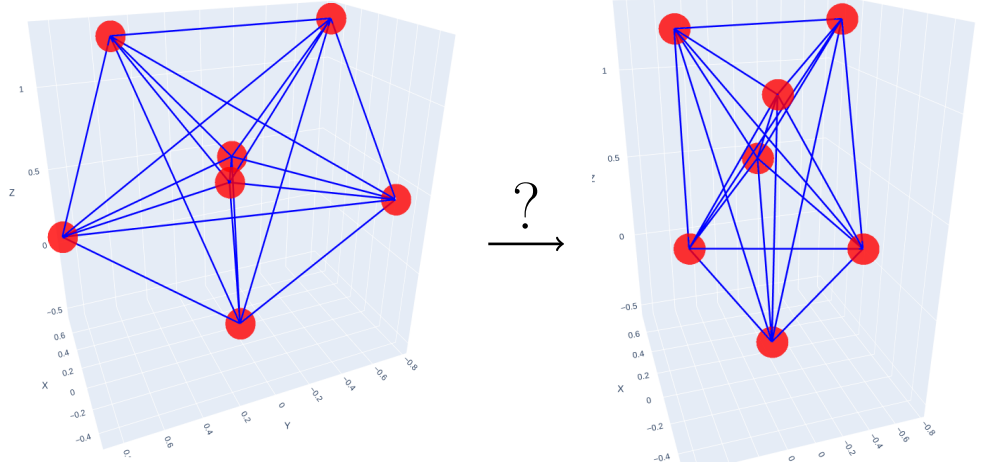


Figure 2.2 Two lowest energy minima of LJ_7 . We wish to find the most probable path from global minimum (left) to second lowest energy minimum (right).

The resulting most probable path is found, results are on Figure 2.3:

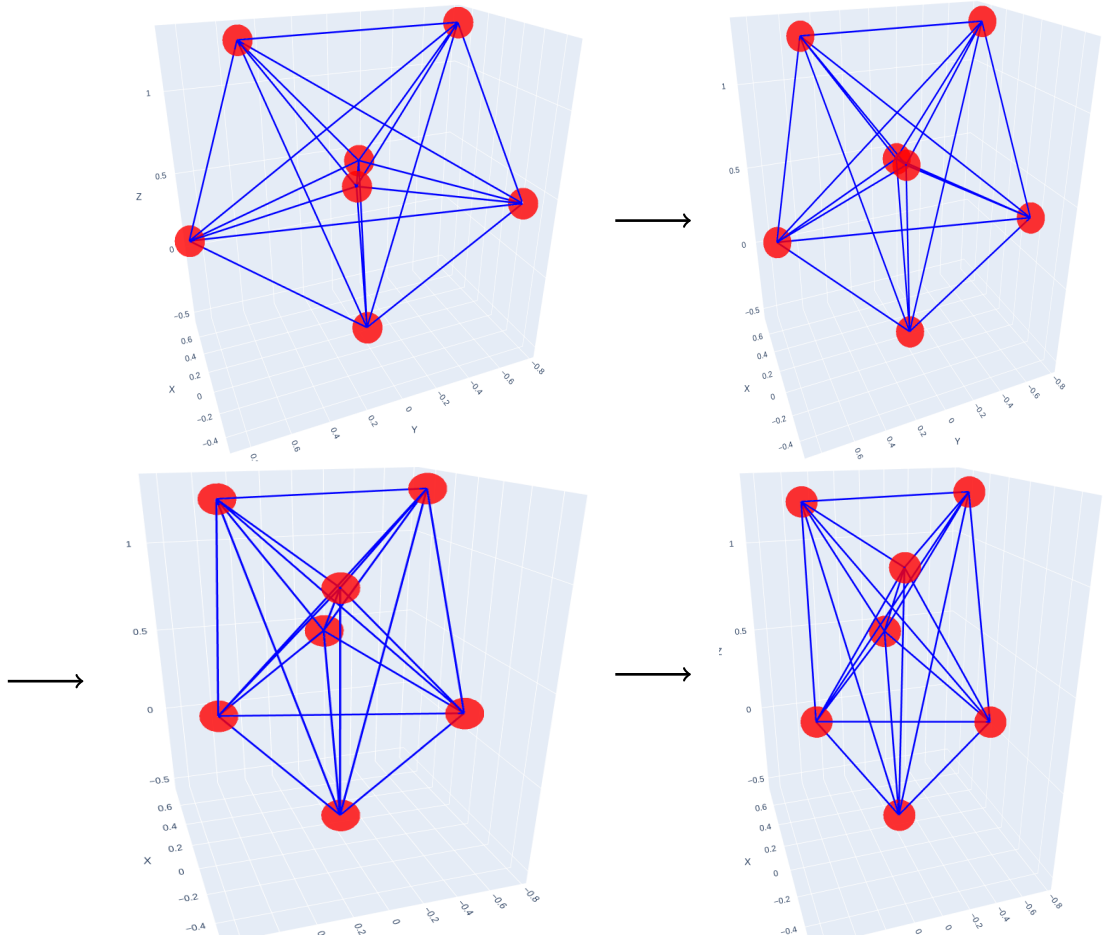


Figure 2.3 path at times 0,0.33,0.66,1 for $\alpha = 5 \cdot 10^{-3}, \beta = -5 \cdot 10^{-4}$. Training converged after 1000 epochs for two hidden layers with 16 neurons, 50 points of path discretization.

Since we have detailed description of the path, we can calculate the free energy profile (and from it activation energy), as is shown on Figure 2.4.

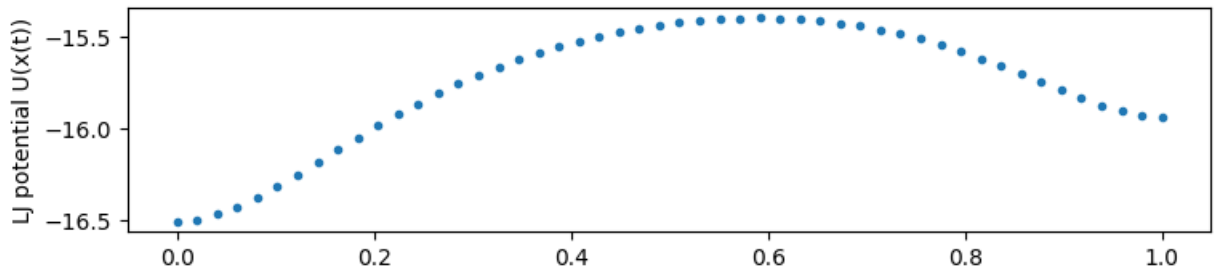


Figure 2.4 Free energy profile $U(\mathbf{x}(t))$, $t \in [0, 1]$. The activation energy is 1.1 (in units of energy where $\epsilon = 1$.) Algorithm converged after 1000 epochs, with two hidden layers of 16 neurons, time discretization to 50 points.

We also check how Hamiltonian is well conserved, as can be seen on Figure 2.5:

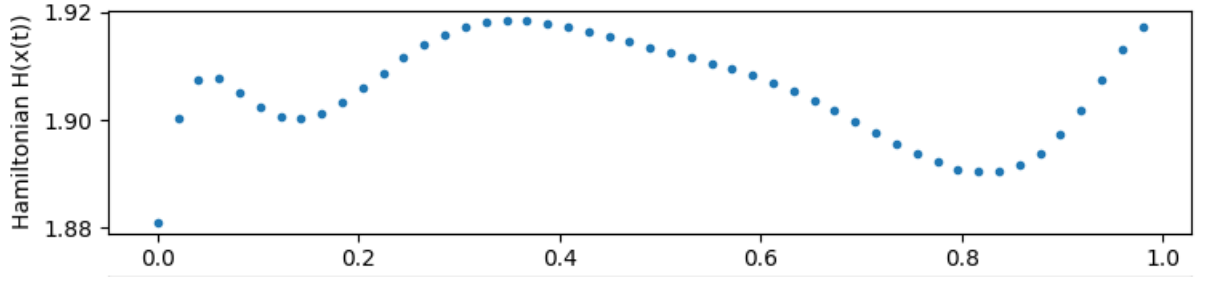


Figure 2.5 Hamiltonian $H(\mathbf{x}(t))$, $t \in [0, 1]$. Deviation $\frac{\sigma_H}{\bar{H}} = 4.765 \cdot 10^{-3}$

Moreover, the deviation of Hamiltonian is smaller for smaller time steps Δt in discretization of S . Lastly we check the distance between the transition point and compare it to a point in the database of saddles from 24. We get that the L^2 error is $\|\mathbf{x}_{transition} - \mathbf{x}_{saddle}\| = 0.0493$, in units of space where $r_0 = 1$.

Remark: For the LJ potential, we ran into the difficulty that one can permute the positions of atoms and still end up with the same cluster, therefore we had to check $\mathbf{x}_{transition}$ against all permutations of all saddle points from 24 and use the Hungarian algorithm 25 to find the permutation minimizing the L^2 distance.

3 Estimating first passage time

There is an apparent defect in our formulation: the fixed time T . We can compute the minimum

$$S^*(T) = \inf_{\substack{\mathbf{x}: [0,T] \rightarrow \mathbb{R}^N \\ \mathbf{x}(0)=\mathbf{x}_0, \mathbf{x}(T)=\mathbf{x}_f}} S_T[\mathbf{x}]$$

for fixed value of T , but for chemical kinetics the interesting quantity is the mean value $\mathbb{E}T$, where T is the first hitting time of transition state/reactant (or equivalently, reaction rate $k = \frac{1}{\mathbb{E}T}$). In this chapter, we discuss this *mean first passage time* (MFPT).

This chapter contains a failed attempt to estimate the MFPT. We are including it because it took us a lot of time to figure out that the approach does not work, and on the numerical results we can at least illustrate the Hamilton Jacobi equations, which might be of interest in it's own right.

3.1 First attempt: $\frac{\partial S}{\partial T} = 0$

The dependence of $S^*(T)$ on T is given by Hamilton-Jacobi equations (see 19 or Appendix B):

$$\frac{\partial S^*(T)}{\partial T} + H(T) = 0 \quad (3.1)$$

where $H(T)$ is the Hamiltonian along the minimizing path. It is tempting to think that $\frac{\partial S^*(T^*)}{\partial T} = 0$ means that the time T^* is the most probable. In the simulations however, we find that $S^*(T)$ is decreasing, as is shown on Figure 3.1

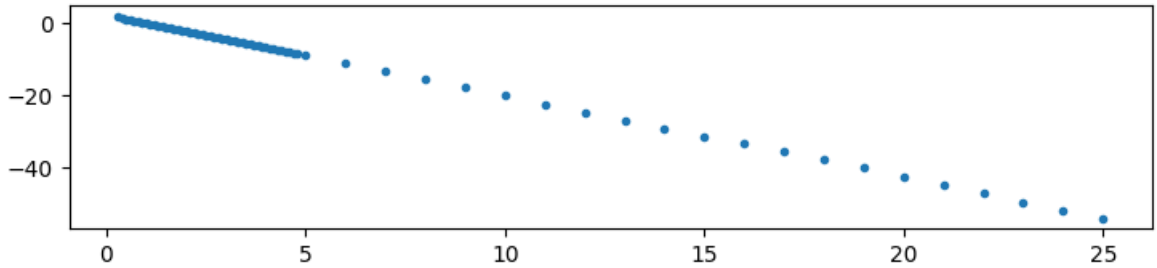


Figure 3.1 dependence of S^* (y axis) on T (x axis)

This is explained by Eq (3.1) when we plot the Hamiltonian, on Figure 3.2:

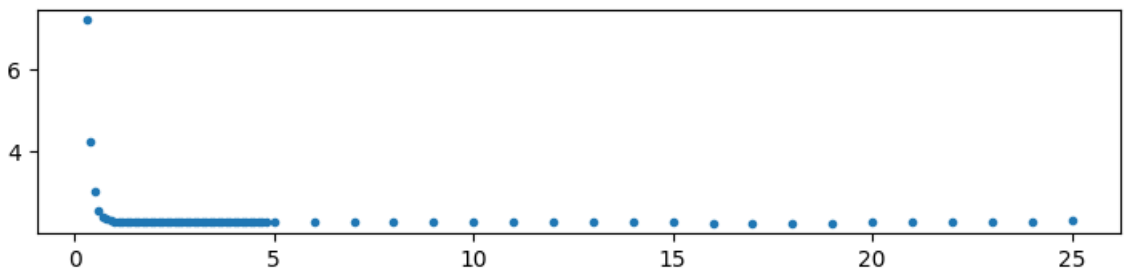


Figure 3.2 Dependence of H (y axis) on T (x axis). $\lim_{T \rightarrow \infty} H(T) \doteq 2.1$

3.2 Second attempt: $\frac{\partial}{\partial T}(S + HT) = 0$

We can heuristically predict asymptotics of $H(T)$ as follows: if the system has a lot of time, it will stay near the minimum $\mathbf{x}_0$ for almost all of it, since there is $-\Delta U$ term in S and $\Delta U(\mathbf{x}_0) > 0$. Therefore in the action will be a term (asymptotically)

$$\int_0^T -\beta \Delta U(\mathbf{x}_0) dt = -H(0)T = -H(T)T$$

and we cannot expect to find a minimum, since $S^* \sim -H(T)T \rightarrow -\infty$. Therefore for $T \rightarrow \infty$ the Hamiltonian approaches a positive limiting value and OM action decreases with no bound. If instead of S^* we work with regularized $S^*(T) + H(T)T$, it will no longer be unbounded. Therefore, we might hope that the regularized S will tell us the most probable time, but that still does not work. There is a minimum, but it does not correspond to the most probable time. For details see 26.

Dependence of $S^*(T) + H(T)T$ can be seen on Figure 3.3:

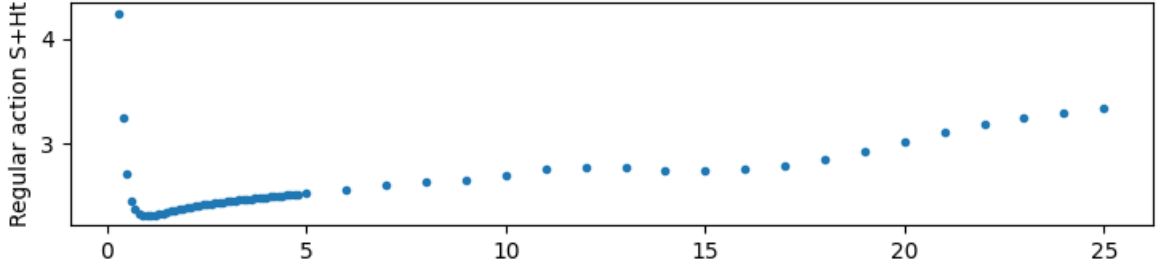


Figure 3.3 $S^*(T) + H(T)T$ depending on T . There is a minimum at $T \doteq 1.1$, which however is not the correct MFPT.

Conclusion

We formulated a method to find the most probable path in a potential landscape and calculated the activation energy and transition states. Our method differs from the standard ones by encoding a continuous path, rather than a sequence of points, which greatly simplifies the method. We checked that the minimizer satisfies a version of conservation of energy (constancy of Hamiltonian). The method was implemented in PyTorch.

We looked at the Müller Brown potential, an analytical potential in 2D. We showed how the most probable path depends on temperature. Then we looked at the Lennard Jones potential for 7 atom cluster and showed that the most probable path gets very close to the well-known saddle point. Then we discussed a failed attempt of estimating the MFPT, which reveals asymptotics of the most probable path as we increase the total time $T \rightarrow \infty$.

There are two main direction we could take this work in the future: the first is consider high-dimensional systems like proteins, and the second is to find an effective method for calculating the MFPT based on the OM action.

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A Derivation of the Onsager Machlup action

Here we lay out the theory of OM action more precisely. Section A.1 is based on 27 chapters 2-5 and sections A.2, A.3 are based on 10.

A.1 Stochastic differential equations

The purpose of this section is to make sense of

$$d\mathbf{x}_t = -\zeta^{-1}\nabla U(\mathbf{x})dt + \sqrt{2D}d\mathbf{W}_t.$$

The term $\mathbf{W}_t$ is the Wiener process, a model of Brownian motion. It represents movement of a particle constantly colliding with smaller particles. The collisions at different times are independent and cause the particle to wiggle very rapidly.

Definition 1 (1-D Wiener process). *The Wiener process $W = (W_t: \Omega \rightarrow \mathbb{R})_{t \in [0, \infty)}$ is a process such that*

$$W_0 = 0$$

$$W_{t_n} - W_{t_{n-1}} \perp W_{t_{n-1}} - W_{t_{n-2}} \perp \dots \perp W_{t_2} - W_{t_1} \quad \forall t_n \geq t_{n-1} \geq \dots \geq t_1$$

$$W_t - W_s \sim \mathcal{N}(0, t - s) \quad \forall t \geq s$$

$W_\bullet$ has continuous trajectory almost surely

From the definition it follows that

$$\begin{pmatrix} W_{t_1} \\ W_{t_2} \\ W_{t_3} \\ \vdots \\ W_{t_n} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} t_1 & t_1 & t_1 & \dots & t_1 \\ t_1 & t_2 & t_2 & \dots & t_2 \\ t_1 & t_2 & t_3 & \dots & t_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & t_3 & \dots & t_n \end{pmatrix} \right) = \mathcal{N}(\vec{0}, C_n)$$

The measure space Ω from Definition 1 is the space of all trajectories $\mathbb{R}^{[0, \infty)}$ and a Wiener measure $\mathcal{D}x$ is constructed from Daniell-Kolmogorov theorem by setting

$$\int_A \mathcal{D}x := \int_{(a_1, b_1) \times \dots \times (a_n, b_n)} \frac{1}{(2\pi)^{\frac{n}{2}} \det(C_n)^{\frac{1}{2}}} \exp\left(-\frac{1}{2}\mathbf{x}^T C_n \mathbf{x}\right) d\mathbf{x}$$

for $A = \{x \in \Omega : x(t_1) \in (a_1, b_1), \dots, x(t_n) \in (a_n, b_n)\}$.

The probability theorist would then write

$$\mathbb{E}[W_{t_1} \in (a_1, b_1), \dots, W_{t_n} \in (a_n, b_n)] := \int_A \mathcal{D}x$$

where $\mathbb{E}$ is expectation with respect to $\mathcal{D}x$. We will call integrals with respect to $\mathcal{D}x$ *path integrals*, since the integration is over space of functions (paths).

Definition 2 (N-D Wiener process). *The N-dimensional Wiener process is $\mathbf{W} = (W^{(1)}, \dots, W^{(N)})$ where $W^{(i)}$ are 1-D Wiener processes such that at any two times $t, s \geq 0$, $W_t^{(i)}$ is independent of $W_s^{(j)}$ for $i \neq j$.*

Definition 3 (Itô integral). Let $X = (X_t)$ be a process. Then the Itô integral of X w.r.t. W is

$$\int_0^T X_t dW_t := \lim_{k \rightarrow \infty} \sum_{i=0}^{N_k-1} X_{t_i^k} (W_{t_{i+1}^k} - W_{t_i^k})$$

where $\mathcal{P}_k = \{t_0^k, \dots, t_{N_k-1}^k\}$ is a sequence of partitions of $[0, T]$ with norm $|\mathcal{P}_k| \rightarrow 0$.

Definition 4 (Stochastic differential equation). We say that a process $\mathbf{x} = (\mathbf{x}_t)_{t \in [0, \infty)}$ solves the SDE

$$d\mathbf{x}_t = \mu(t, \mathbf{x}_t)dt + \sigma(t, \mathbf{x}_t)d\mathbf{W}_t$$

if for all $t \in [0, T]$ it holds

$$\mathbf{x}_t = \mathbf{x}_0 + \int_0^t \mu(s, \mathbf{x}_s)ds + \int_0^t \sigma(s, \mathbf{x}_s)d\mathbf{W}_s$$

where the first integral is classical (Lebesgue) and the second integral is Itô.

Theorem 2 (Itô formula). Let $f = f(\mathbf{x})$ be function of space and let the process $\mathbf{x}$ solve

$$d\mathbf{x}_t = \mu(t, \mathbf{x}_t)dt + \sigma(t, \mathbf{x}_t)d\mathbf{W}_t$$

Consider $f(\mathbf{x}_t)$, which is a process derived from $\mathbf{x}_t$. Then it holds

$$\begin{aligned} df(\mathbf{x}_t) &= \mu \nabla f \cdot d\mathbf{x}_t + \frac{1}{2} \sigma^2 \Delta f dt \\ &= \left(\mu \nabla f \cdot + \frac{1}{2} \sigma^2 \Delta f \right) dt + \sigma \nabla f \cdot d\mathbf{W}_t \end{aligned}$$

A.2 Langevin $\rightarrow$ Fokker Planck equation

Next, we calculate the probability density $\Pi(\mathbf{x}, t)$ that a particle starting at $\mathbf{x}_0$ at time 0 will end at $\mathbf{x}$ at time t . It will be given by the Fokker Planck equation:

$$\begin{aligned} \partial_t \Pi - \nabla \cdot \left(\Pi \frac{\nabla U}{\zeta} \right) - D \Delta \Pi &= 0 \\ \Pi(\mathbf{x}, 0) &= \delta(\mathbf{x} - \mathbf{x}_0) \end{aligned} \tag{A.1}$$

For simplicity let

$$d\mathbf{x} = \mu dt + \sigma d\mathbf{W},$$

where $\mu = -\zeta^{-1} \nabla U(\mathbf{x})$, $\sigma = \sqrt{2D}$. Let $f(\mathbf{x})$ be a test function. Denote

$$\langle f \rangle(t) := \mathbb{E} f(\mathbf{x}_t) = \int f(\mathbf{y}) \Pi(t, \mathbf{y}) d\mathbf{y}$$

On one hand

$$\frac{d}{dt} \langle f \rangle = \int f(\mathbf{y}) \partial_t \Pi(t, \mathbf{y}) d\mathbf{y}.$$

On the other hand, by averaging Itô's formula, since $\langle d\mathbf{W} \rangle = 0$

$$\begin{aligned} df(\mathbf{x}_t) &= \left(\nabla f \cdot \mu + \frac{1}{2} \Delta f \sigma^2 \right) dt + \sigma \nabla f \cdot d\mathbf{W} \\ \implies \frac{d}{dt} \langle f \rangle &= \nabla f \cdot \mu + \frac{1}{2} \Delta f \sigma^2 \end{aligned}$$

So

$$\forall f : \int f \partial_t \Pi d\mathbf{x} = \int \Pi \left(\nabla f \cdot \mu + \frac{1}{2} \Delta f \sigma^2 \right) d\mathbf{x}$$

but this is just Eq (A.1) in distributions.

A.3 Fokker Planck equation $\rightarrow$ OM action

Now we show the solution of Eq (A.1) can be written as the path integral

$$\begin{aligned} \Pi[\mathbf{x}_f, t] &= \int_{\mathbf{x}_0, 0}^{\mathbf{x}_f, t} \exp \left(-\frac{S[\mathbf{x}(t)]}{2D} \right) \mathcal{D}\mathbf{x}(t), \\ S[\mathbf{x}(t)] &= \frac{[U]_{\mathbf{x}_0}^{\mathbf{x}_f}}{\zeta} + \frac{1}{2} \int_0^t \left(|\dot{\mathbf{x}}|^2 + \left(\frac{\nabla U}{\zeta} \right)^2 - \frac{2D}{\zeta} \Delta U \right) d\tau \end{aligned}$$

STEP 1: transformation to damped heat equation.

$$\Pi[\mathbf{x}, t] = G[\mathbf{x}(t)] \exp \left(\frac{\phi(\mathbf{x}_0) - \phi(\mathbf{x})}{2D} \right), \text{ where } \phi = \frac{U}{\zeta}, \text{ t.j.} \mu = -\nabla \phi$$

We write $\Pi = G \exp(\dots)$, derivatives become

$$\begin{aligned} \partial_t \Pi &= (\partial_t G) \exp(\dots) \\ \nabla \cdot (\Pi \mu) &= \exp(\dots) \nabla G \cdot \mu + G \frac{1}{2D} \exp(\dots) (-\nabla \phi) \cdot \mu + G \exp(\dots) \nabla \cdot \mu \\ &= \left(-\nabla G \cdot \nabla \phi + \frac{G}{2D} (\nabla \phi)^2 - G \Delta \phi \right) \exp(\dots) \\ \Delta \Pi &= (\Delta G) \exp(\dots) + 2 \nabla G \cdot \left(-\frac{1}{2D} \nabla \phi \exp(\dots) \right) + G \Delta (\exp(\dots)) \\ &= \left(\Delta G - \frac{1}{D} \nabla G \cdot \nabla \phi - \frac{G}{2D} \Delta \phi + \frac{G}{4D^2} (\nabla \phi)^2 \right) \exp(\dots) \end{aligned}$$

After using Fokker Planck, we are left with

$$\partial_t G = -kG + D\Delta G, \quad k(\mathbf{x}) = -\frac{1}{4D} (\nabla \phi(\mathbf{x}))^2 + \frac{1}{2} \Delta \phi(\mathbf{x})$$

STEP 2: solving the damped heat equation over one small time step.

We want to know the probability $\Pi[\mathbf{x}_{j+1}, t_{j+1}; \mathbf{x}_j, t_j]$ that the process beginning at $\mathbf{x}_j$ in time t_j ends up at $\mathbf{x}_{j+1}$ in time $t_{j+1} = t_j + \Delta t$. By stationarity, we can translate time:

$$G[\mathbf{x}_{j+1}, t_{j+1}; \mathbf{x}_j, t_j] = G[\mathbf{x}_{j+1}, \Delta t; \mathbf{x}_j, 0] =: G(\mathbf{x}_{j+1})$$

Function $G = G(\mathbf{x}_{j+1})$ solves $\partial_t G = -kG + D\Delta G$. For Δt small enough, $k(\mathbf{x}(t)) \doteq k(\mathbf{x}(t_{j+1}))$ is constant for $t \in [t_j, t_{j+1}]$:

$$\begin{aligned} e^{kt} \partial_t G &= -e^{kt} kG + D e^{kt} \Delta G \\ \partial_t (e^{kt} G) &= D \Delta (e^{kt} G) \end{aligned} \tag{A.2}$$

By solving Eq (A.2) we get

$$\begin{aligned}
G[\mathbf{x}_{j+1}, t_{j+1}; \mathbf{x}_j, t_j] &= \\
G(\mathbf{x}_{j+1}) &= (4\pi D \Delta t)^{-\frac{3}{2}} \exp(-k(\mathbf{x}_{j+1}) \Delta t) \exp\left(-\frac{|\mathbf{x}_{j+1} - \mathbf{x}_j|^2}{4D \Delta t}\right) \\
\Pi[\mathbf{x}_{j+1}, t_{j+1}; \mathbf{x}_j, t_j] &= \\
(4\pi D \Delta t)^{-\frac{3}{2}} \exp\left(-k(\mathbf{x}_{j+1}) \Delta t - \frac{|\mathbf{x}_{j+1} - \mathbf{x}_j|^2}{4D \Delta t} + \frac{\phi(\mathbf{x}_j) - \phi(\mathbf{x}_{j+1})}{2D}\right)
\end{aligned}$$

STEP 3: discrete approximation of path.

For $N \in \mathbb{N}$, $\mathbf{x}(t)$ passes through $(\mathbf{x}_j, t_j)$, $j = 0, 1, \dots, N$, $\mathbf{x}_j = \mathbf{x}(t_j)$, $t_{j+1} = t_j + \Delta t$, $\Delta t = \frac{T}{N}$. By markov property of $\mathbf{x}$, we can split the probability of this sequence into product of probabilities:

$$\begin{aligned}
\Pi_N &:= \Pi[\mathbf{x}_N, t_N; \dots; \mathbf{x}_1, t_1; \mathbf{x}_0, 0] = \prod_{j=0}^{N-1} \Pi[\mathbf{x}_{j+1}, t_{j+1}; \mathbf{x}_j, t_j] \\
&= (4\pi D \Delta t)^{-\frac{3}{2}N} \exp\left(\sum_{j=0}^{N-1} -k(\mathbf{x}_{j+1}) \Delta t - \frac{|\mathbf{x}_{j+1} - \mathbf{x}_j|^2}{4D \Delta t} + \frac{\phi(\mathbf{x}_j) - \phi(\mathbf{x}_{j+1})}{2D}\right) \\
&\doteq (4\pi D \Delta t)^{-\frac{3}{2}N} \exp\left(\sum_{j=0}^{N-1} \left(-k(\mathbf{x}_{j+1}) - \frac{|\dot{\mathbf{x}}(t_{j+1})|^2}{4D} - \frac{\nabla \phi(\mathbf{x}_{j+1}) \cdot \dot{\mathbf{x}}(t_{j+1})}{2D}\right) \Delta t\right).
\end{aligned}$$

STEP 4: limit $N \rightarrow \infty$.

In the limit, we get Riemann integrals over $[0, T]$ and a path integral over all paths $\mathbf{x}: [0, T] \rightarrow \mathbb{R}^N$ such that $\mathbf{x}(0) = \mathbf{x}_0$, $\mathbf{x}(T) = \mathbf{x}_f$:

$$\begin{aligned}
\sum_{j=0}^{N-1} f(t_{j+1}) \Delta t &\rightarrow \int_0^t f(t) dt \\
(4\pi D \Delta t)^{-\frac{3}{2}N} \underbrace{\int \dots \int_N g[\mathbf{x}(t)] d\mathbf{x}_1 \dots d\mathbf{x}_N}_{N} &\rightarrow \int_{\mathbf{x}_0, 0}^{\mathbf{x}_f, t} g[\mathbf{x}(t)] \mathcal{D}\mathbf{x}(t)
\end{aligned}$$

Finally:

$$\begin{aligned}
\Pi[\mathbf{x}, t; \mathbf{x}_0, 0] &= \lim_{N \rightarrow \infty} \left(\int \dots \int \Pi_N d\mathbf{x}_1 \dots d\mathbf{x}_N \right) \\
&= \int_{\mathbf{x}_0, 0}^{\mathbf{x}, t} \exp\left(-\int_0^t \left(k(\mathbf{x}(t)) + \frac{1}{4D} |\dot{\mathbf{x}}|^2 + \frac{1}{2D} \nabla \phi(\mathbf{x}(t)) \cdot \dot{\mathbf{x}}(t)\right) d\tau\right) \mathcal{D}\mathbf{x}(t) \\
&= \int_{\mathbf{x}_0, 0}^{\mathbf{x}, t} \exp\left(-\frac{1}{4D} \int_0^t \left((\nabla \phi(\mathbf{x}))^2 - 2D \Delta \phi(\mathbf{x}) + |\dot{\mathbf{x}}|^2\right) d\tau - \frac{\int_0^t \nabla \phi(\mathbf{x}(t)) \cdot d\mathbf{x}}{2D}\right) \mathcal{D}\mathbf{x}(t) \\
&= \int_{\mathbf{x}_0, 0}^{\mathbf{x}, t} \exp\left(-\frac{1}{4D} \int_0^t \left(\left(\frac{\nabla U(\mathbf{x})}{\zeta}\right)^2 - \frac{2D}{\zeta} \Delta U(\mathbf{x}) + |\dot{\mathbf{x}}|^2\right) d\tau - \frac{[U]_{\mathbf{x}_0}^{\mathbf{x}_f}}{2D\zeta}\right) \mathcal{D}\mathbf{x}(t) \\
&= \int_{\mathbf{x}_0, 0}^{\mathbf{x}, t} \exp\left(-\frac{S[\mathbf{x}(t)]}{2D}\right) \mathcal{D}\mathbf{x}(t).
\end{aligned}$$

B Derivation of Hamilton-Jacobi equations

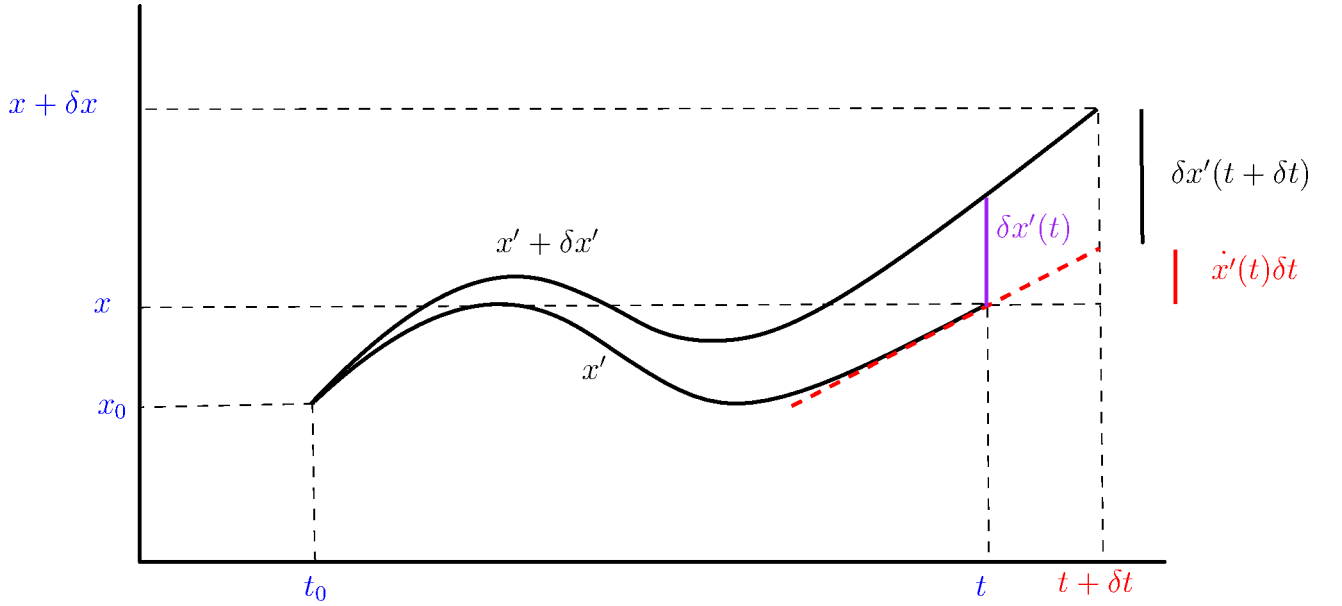
This section is based on 19.

Suppose for every possible starting time and position $(t_0, \mathbf{x}_0)$ and final time and position $(t, \mathbf{x})$ we find the minimum of S , and want to see how it behaves as a function of $t_0, \mathbf{x}_0, t, \mathbf{x}$. This is Hamilton's principle function:

$$S(\mathbf{x}, t, \mathbf{x}_0, t_0) := \inf_{\substack{\mathbf{x}': [t_0, t] \rightarrow \mathbb{R}^N \\ \mathbf{x}'(t_0) = \mathbf{x}_0, \mathbf{x}'(t) = \mathbf{x}}} S[\mathbf{x}']$$

Hamilton-Jacobi equations show how S changes under small changes of $\mathbf{x}, t$. Suppose the (minimizer) path connecting $(t_0, \mathbf{x}_0) \rightarrow (t, \mathbf{x})$ is $\mathbf{x}'$ and the path connecting $(t_0, \mathbf{x}_0) \rightarrow (t + \delta t, \mathbf{x} + \delta \mathbf{x})$ is $\mathbf{x}' + \delta \mathbf{x}'$:

$$\begin{aligned} \delta S &= \delta \int_{t_0}^t L(\mathbf{x}', \dot{\mathbf{x}}') dt = \int_{t_0}^t \left(\frac{\partial L}{\partial \mathbf{x}} \delta \mathbf{x}' + \frac{\partial L}{\partial \dot{\mathbf{x}}} \frac{d}{dt} \delta \mathbf{x}' \right) dt + L(\mathbf{x}'(t), \dot{\mathbf{x}}'(t)) \delta t \\ &= \int_{t_0}^t \underbrace{\left(\frac{\partial L}{\partial \mathbf{x}} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{x}}} \right)}_{=0} \delta \mathbf{x}' dt + \left[\frac{\partial L}{\partial \dot{\mathbf{x}}} \delta \mathbf{x}' \right]_{t_0}^t + L(\mathbf{x}'(t), \dot{\mathbf{x}}'(t)) \delta t = \frac{\partial L}{\partial \dot{\mathbf{x}}} \Big|_t \delta \mathbf{x}'(t) + L|_t \delta t \end{aligned}$$



We need to compare the positions of $\mathbf{x}'$ and $\mathbf{x}' + \delta \mathbf{x}'$ at time t , however this is not the same as $\delta \mathbf{x}$, because $\mathbf{x}' + \delta \mathbf{x}'$ has extra time δt to move a bit further:

$$\delta \mathbf{x}'(t) \approx \delta \mathbf{x}'(t + \delta t) = \delta \mathbf{x} - \dot{\mathbf{x}}'(t) \delta t$$

Therefore (in terms of Hamiltonian variables $\mathbf{x}, \mathbf{p}, H$)

$$\delta S = \frac{\partial L}{\partial \dot{\mathbf{x}}} \delta \mathbf{x} + \left(L - \frac{\partial L}{\partial \dot{\mathbf{x}}} \dot{\mathbf{x}} \right) \delta t \Rightarrow \frac{\partial S}{\partial \mathbf{x}} = \mathbf{p}, \quad \frac{\partial S}{\partial t} = -H$$